

UNIT I: RELATIONAL MODEL & RELATIONAL ALGEBRA OPERATORS

Weightage: 25% | Syllabus Section 1

1. Relational Algebra Primitives

Selection (σ): Rows meeting condition: $\sigma_p(R)$.

Commutative: $\sigma_c(\sigma_d(R)) = \sigma_d(\sigma_c(R))$.

Projection (π): Columns selected (duplicates purged):

$\pi_{A_1, \dots, A_k}(R)$.

Cartesian Product ($\times$): $R \times S$. Degree = $\deg(R) + \deg(S)$; Cardinality = $|R| \times |S|$.

Natural Join ($\bowtie$): $R \bowtie S = \pi_{\text{Attrs}}(\sigma_{R.A=S.A}(R \times S))$.

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2. Functional Dependencies & Closures

FD Definition: $X \rightarrow Y$ means identical X values strictly imply identical Y values.

Armstrong's Axioms:

• Reflexivity: If $Y \subseteq X \implies X \rightarrow Y$.

• Augmentation: $X \rightarrow Y \implies XZ \rightarrow YZ$.

• Transitivity: $X \rightarrow Y \wedge Y \rightarrow Z \implies X \rightarrow Z$.

Attribute Closure (X^+): Set of all attributes functionally determined by X . If X^+ contains all attributes, X is a

****Superkey**!**

3. Canonical Cover & Minimal Basis

Minimal Cover Steps:

1. Split right-hand sides into singletons ($X \rightarrow YZ \implies X \rightarrow Y, X \rightarrow Z$).

2. Remove extraneous left-hand side attributes.

3. Eliminate redundant functional dependencies using attribute closure.

UNIT II: DATABASE NORMALIZATION (1NF TO BCNF)

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1. Normal Forms Hierarchy & Definitions

Normal Form	Strict Mathematical Condition	Anomalies Prevented
1NF	All attribute values are atomic; no repeating groups.	Nested relations
2NF	In 1NF and NO non-prime attribute is partially dependent on any candidate key ($\neg(\text{ProperSubKey} \rightarrow \text{NonPrime})$).	Partial Update anomalies
3NF	For every non-trivial FD $X \rightarrow A$: either X is a Superkey OR A is a prime attribute.	Transitive Dependencies
BCNF	For EVERY non-trivial FD $X \rightarrow A$: X MUST be a Superkey! Strictly no non-superkey determinants.	Redundancy in candidate keys

2. Lossless Join & Dependency Preservation

Lossless Decomposition Test: Decomposition of R into R_1, R_2 is lossless if and only if: $(R_1 \cap R_2) \rightarrow R_1$ OR $(R_1 \cap R_2) \rightarrow R_2$.

(Common attribute must be a candidate key in at least one decomposed sub-relation!)

Dependency Preservation: $(F_1 \cup F_2)^+ = F^+$. Note: BCNF decomposition is always lossless, but may NOT always preserve functional dependencies (3NF always preserves dependencies!).

UNIT III: TRANSACTION PROCESSING & CONCURRENCY CONTROL

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1. ACID Properties & States

Atomicity: All or nothing (Rollback via UNDO log).
Consistency: Preserves database integrity constraints.
Isolation: Concurrently executing transactions cannot see intermediate uncommitted states.
Durability: Committed updates persist across crashes (REDO log).

2. Conflict Serializability

Conflicting Operations: Two operations on same data item, by different transactions, where at least one is a WRITE.
Precedence Graph Testing: Directed edge $T_i \rightarrow T_j$ if T_i performs conflicting op before T_j .
• Graph is **Acyclic** $\iff$ Schedule is **Conflict Serializable**!
• Equivalent serial schedule obtained via **Topological Sort**.

3. Two-Phase Locking (2PL)

Growing Phase: Locks acquired; none released.
Shrinking Phase: Locks released; none acquired.
• *Theorem:* 2PL guarantees conflict serializability!
• *Strict 2PL:* Hold all exclusive (X) locks until commit (prevents cascading rollbacks).
• *Rigorous 2PL:* Hold all locks until commit.

UNIT IV: INDEXING, B+ TREES & QUERY OPTIMIZATION

Weightage: 25% | Syllabus Section 4

1. B+ Tree Indexing Formulas

B+ Tree Structure: All keys stored in leaf nodes linked as linked list. Internal nodes serve only as search routers.
Order p of Internal Node: $p \cdot P_{\text{block}} + (p - 1) \cdot K \leq B$, where B is block size, K key size, P_{block} pointer size.
Search Complexity: Height $h \leq \lceil \log_{\lceil p/2 \rceil} N \rceil$. Disk I/O is logarithmic, provably bounding worst-case latency to 3–4 block reads!

2. Query Cost & Optimization

Heuristics: Perform σ (selections) and π (projections) as early as possible to minimize intermediate relation size before computing expensive $\bowtie$ (joins).
Block Nested Loop Join: Cost $B_R + \lceil \frac{B_R}{M-2} \rceil \cdot B_S$, where M is buffer memory pages.